

PHASE 3 — POST-QUANTUM CRYPTOGRAPHY (PQC)

The Cryptographic Transition Beyond RSA and ECC

Project: Enterprise Cryptographic Discovery & Analysis Tool (ECDAT)

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1. Purpose of This Phase

Phase 2 established the problem:

Quantum computers threaten important cryptographic systems currently deployed throughout the Internet and enterprise infrastructure.

Phase 3 addresses the next question:

What should organizations use instead?

The answer is:

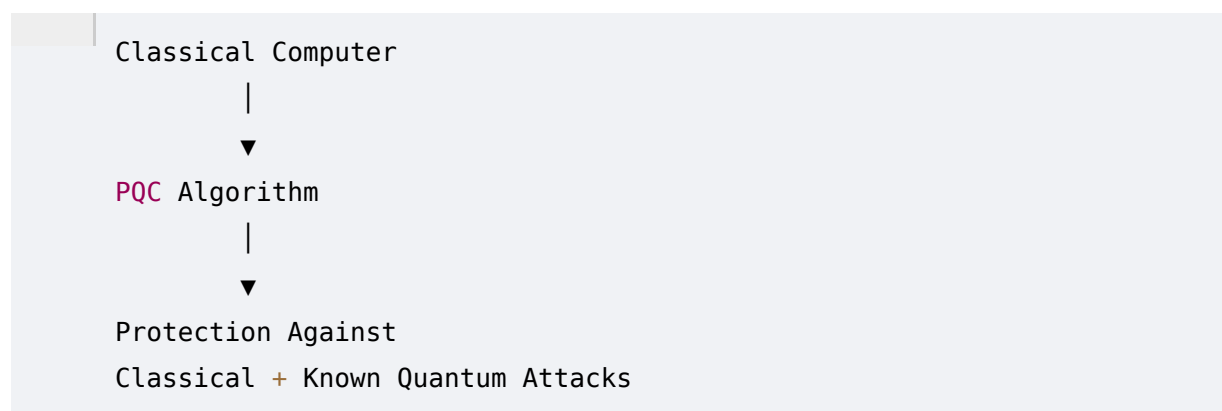
Post-Quantum Cryptography

Post-Quantum Cryptography, commonly abbreviated **PQC**, refers to cryptographic algorithms designed to remain secure against attackers possessing quantum computers, while running on conventional computing infrastructure.

This distinction is important.

PQC does **not** require a quantum computer.

Instead:

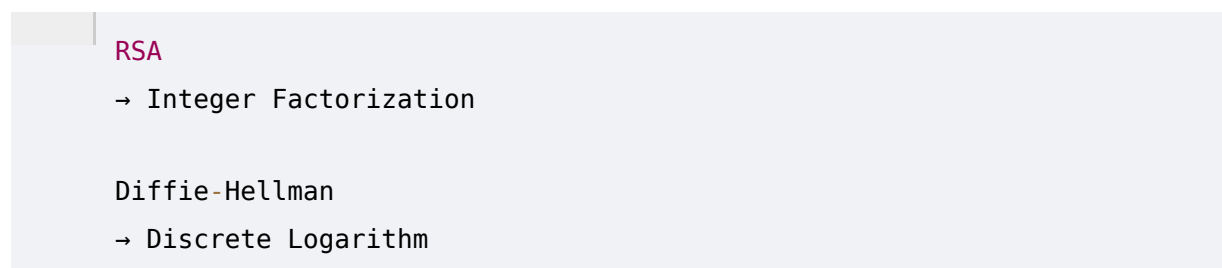


This makes PQC especially attractive for large-scale enterprise migration.

2. What Is Post-Quantum Cryptography?

PQC is a family of cryptographic techniques designed to address cryptanalytic threats from sufficiently powerful quantum computers.

Traditional public-key cryptography relies heavily on mathematical problems such as:



ECC

→ Elliptic Curve Discrete Logarithm

Shor's algorithm threatens these assumptions.

PQC attempts to replace them with mathematical problems for which no efficient classical or quantum attacks are currently known.

Examples include problems involving:

- Lattices
- Error-correcting codes
- Hash functions

3. Why PQC Is Necessary

Consider a company using:

RSA-2048

ECDSA P-256

ECDH P-256

Today these algorithms are widely deployed and remain important components of modern infrastructure.

However, they have a fundamental long-term problem:

Cryptographically Relevant Quantum Computer

↓

Shor's Algorithm

↓

Public-Key Cryptography

↓

Security Assumptions Fail

Replacing this infrastructure can take years.

Large organizations may have:

- Millions of certificates
- Legacy applications
- Embedded devices
- Hardware dependencies

- Third-party software
- Custom protocols
- Long-lived data

Therefore, migration must begin before the threat becomes operational.

4. PQC vs Quantum Cryptography

These terms are frequently confused.

Post-Quantum Cryptography

PQC:

- Runs on conventional computers.
 - Uses mathematical algorithms.
 - Is intended to resist quantum attacks.
 - Can potentially be integrated into existing networks and software.
-

Quantum Cryptography

Quantum cryptography uses actual quantum communication phenomena.

A famous example is:

Quantum Key Distribution (QKD).

QKD requires specialized quantum communication infrastructure.

Therefore:

```
PQC
=
Classical Infrastructure
+
Quantum-Resistant Algorithms
```

while:

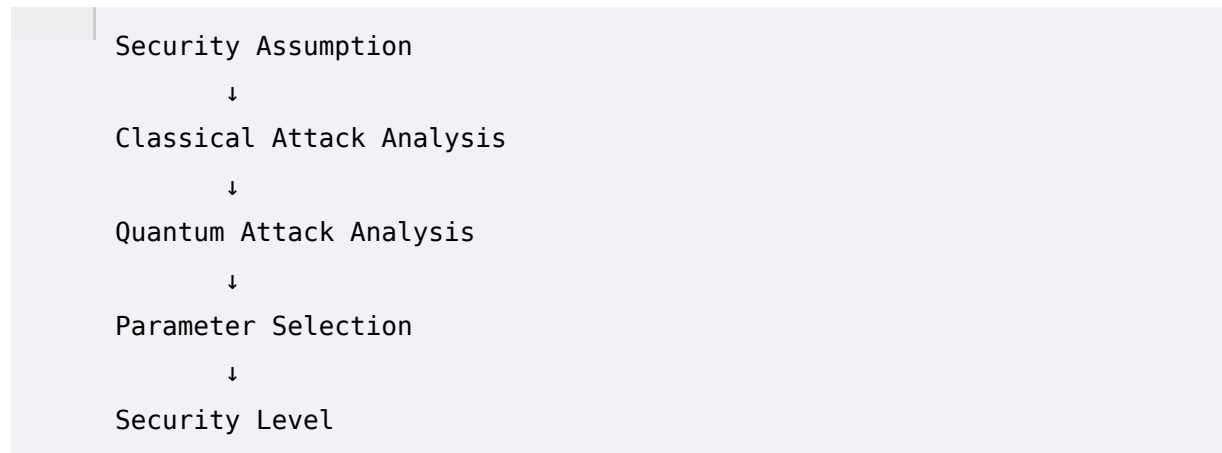
```
QKD
=
Quantum Communication Infrastructure
```

ECDAT is primarily concerned with **PQC migration**, not replacing enterprise infrastructure with QKD.

5. What Makes a Cryptographic Algorithm Post-Quantum?

There is no mathematical label that guarantees an algorithm is permanently secure. Instead, PQC algorithms are designed around computational problems believed to remain difficult even for quantum computers.

A useful conceptual model is:



The cryptographic community continually evaluates these assumptions.

Therefore:

Post-quantum security is based on current knowledge and security analysis, not an absolute guarantee about every future algorithmic breakthrough.

6. The NIST PQC Standardization Effort

The U.S. National Institute of Standards and Technology (**NIST**) conducted a major multi-year process to evaluate candidate post-quantum algorithms.

The process included:

- Public submissions
- Cryptanalysis
- Security analysis
- Performance evaluation
- Implementation analysis
- Multiple selection rounds

The goal was to establish standardized algorithms suitable for practical deployment.

The resulting standards are especially important to ECDAT because the NTRO problem statement explicitly asks the system to:

Recommend suitable alternatives (PQC/Hybrid algorithms).

Therefore, ECDAT should maintain an up-to-date knowledge base of standardized PQC algorithms and their intended cryptographic roles.

7. The Main PQC Families

PQC research has explored several mathematical families.

The most important categories include:

```
graph TD; A[Post-Quantum Cryptography] --> B[Lattice-Based]; A --> C[Hash-Based]; A --> D[Code-Based]; A --> E[Multivariate]; A --> F[Other Specialized Constructions];
```

Some candidate families have experienced major cryptanalytic failures.

This illustrates an important lesson:

PQC algorithms must be subjected to extensive public cryptanalysis before they can be trusted.

8. Lattice-Based Cryptography

Lattice-based cryptography is one of the most important PQC approaches.

A lattice is a mathematical structure generated from combinations of basis vectors.

Simplified:

```
graph LR; v1 --> A[ ]; A --> B[ ];
```

Combining lattice vectors produces a structured set of points.

Certain computational problems on lattices are believed to be hard even for quantum computers.

Important problems include:

- Learning With Errors (LWE)
- Module Learning With Errors (Module-LWE)
- Shortest Vector Problem (SVP)
- Closest Vector Problem (CVP)

Modern PQC standards use structured lattice constructions.

9. Hash-Based Cryptography

Hash-based cryptography relies primarily on the security properties of cryptographic hash functions.

Its strongest practical application is digital signatures.

Advantages:

- Conservative security assumptions
- Relies heavily on well-studied hash functions
- Strong conceptual simplicity

Disadvantages can include:

- Large signatures
- More complex state management for some constructions
- Performance and operational tradeoffs

Hash-based signatures are particularly valuable as an alternative security foundation.

10. Code-Based Cryptography

Code-based cryptography relies on problems involving error-correcting codes.

The basic concept involves adding errors to encoded information and constructing a cryptographic problem around decoding.

Some code-based schemes have been studied for decades.

Advantages:

- Long research history
- Different mathematical assumptions from lattice systems

Disadvantages:

- Potentially very large keys
- Significant bandwidth and storage implications

This tradeoff is important for ECDAT's recommendation engine.

11. Multivariate Cryptography

Multivariate schemes rely on solving systems of multivariate polynomial equations.

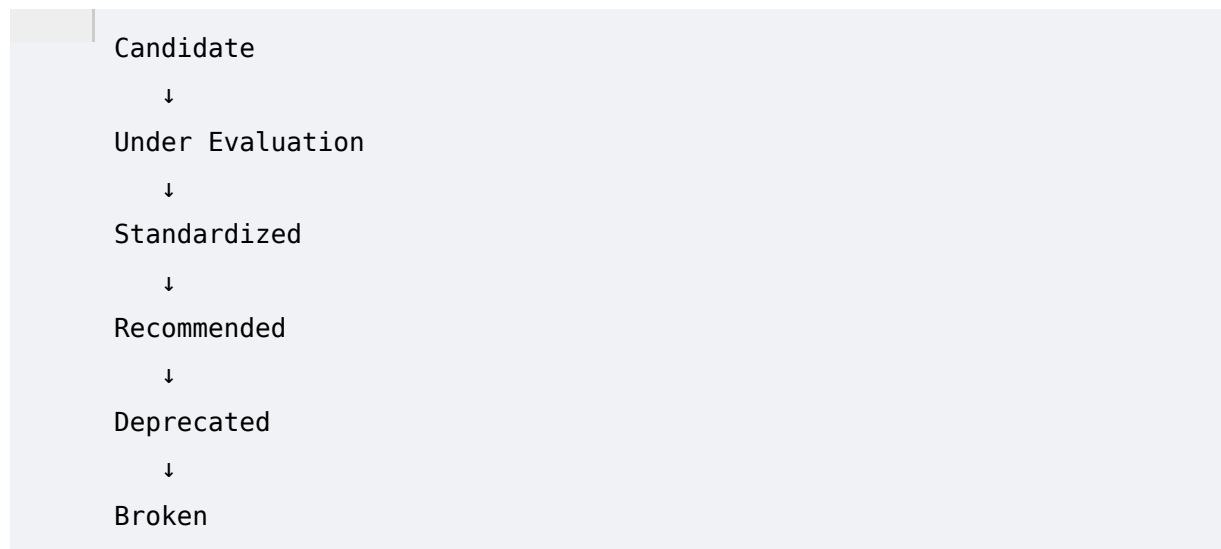
Historically, they were considered a promising PQC family.

However, several proposed schemes have suffered major cryptanalytic breaks.

The lesson is important:

"Post-quantum candidate" does not automatically mean "safe."

ECDAT should therefore maintain an algorithm lifecycle:



12. Isogeny-Based Cryptography

Isogeny-based cryptography was another promising research direction.

It attracted significant attention because of potentially compact key sizes.

However, important candidate systems were later broken by classical cryptanalysis.

This demonstrates why enterprise systems should prioritize standardized and extensively analyzed algorithms rather than experimental research candidates.

13. Key Encapsulation Mechanisms

One of the most important concepts in PQC is the:

Key Encapsulation Mechanism

or **KEM**.

A KEM enables two parties to establish a shared secret over a public channel.

Conceptually:



The resulting shared secret can then be used with symmetric encryption.

This is conceptually similar to the role played by key-exchange mechanisms in classical protocols.

14. Digital Signatures

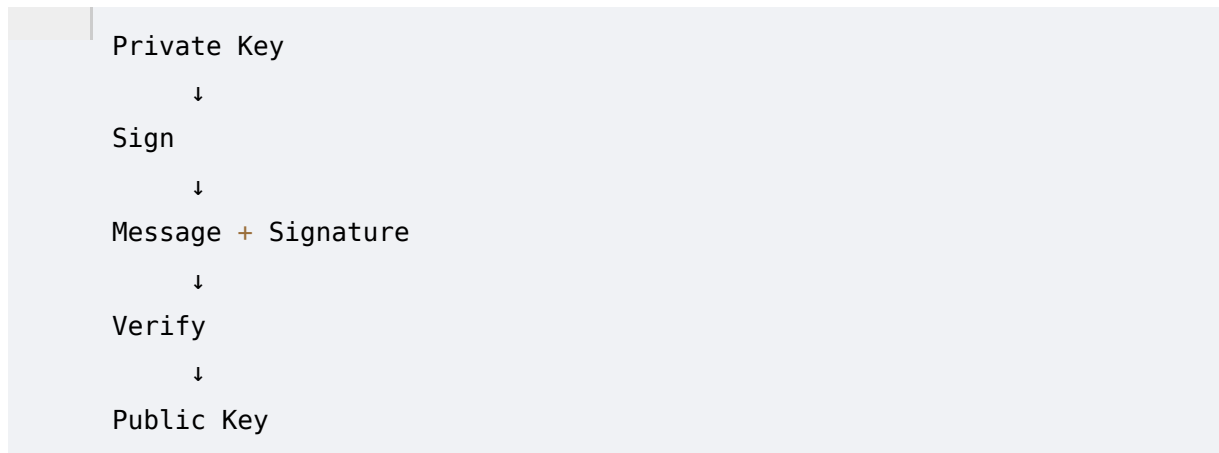
PQC also needs quantum-resistant digital signatures.

A digital signature allows:

- Authentication
- Integrity
- Proof of origin
- Software signing

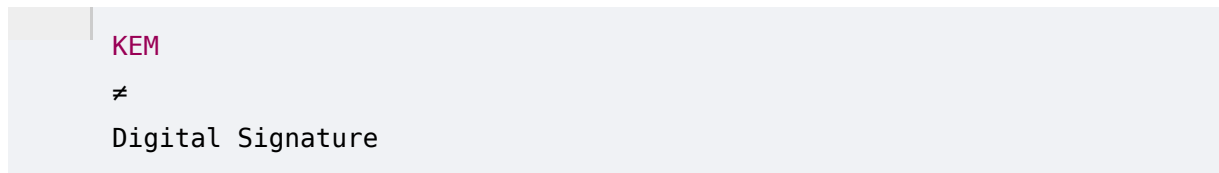
- Certificate signing

A PQC signature system must support:



This is a separate problem from key establishment.

Therefore:



ECDAT must not recommend a KEM as a replacement for a signature algorithm.

15. ML-KEM

One of the most important standardized PQC mechanisms is:

ML-KEM

ML-KEM is a standardized lattice-based key encapsulation mechanism derived from the CRYSTALS-Kyber family.

Its purpose is:

Establish a shared secret securely in the presence of quantum-capable attackers.

ML-KEM is intended for applications such as:

- Key establishment
- Secure communication
- TLS integration
- VPN protocols
- Application-level encryption

ML-KEM Variants

Common parameter sets include:

```
ML-KEM-512
ML-KEM-768
ML-KEM-1024
```

These provide different security/performance tradeoffs.

ECDAT should therefore record the exact parameter set.

For example:

```
Algorithm:
ML-KEM

Variant:
ML-KEM-768

Purpose:
Key Encapsulation

Security Category:
PQC

Usage:
TLS key establishment
```

16. ML-DSA

Another major standardized PQC algorithm is:

ML-DSA

ML-DSA is a lattice-based digital signature standard derived from the CRYSTALS-Dilithium family.

It provides:

- Digital signatures
- Authentication
- Integrity
- Software signing

Potential applications include:

- Code signing
- Certificates
- Firmware signing
- Document signing
- API authentication
- Enterprise identity

ML-DSA Parameter Sets

Common standardized parameter sets include:

```
ML-DSA-44
ML-DSA-65
ML-DSA-87
```

Again, ECDAT must identify the exact parameter set.

17. SLH-DSA

Another standardized PQC signature system is:

SLH-DSA

SLH-DSA is a hash-based digital signature standard derived from the SPHINCS+ family.

Its major conceptual advantage is that its security foundation is based primarily on cryptographic hash functions rather than lattice assumptions.

This provides algorithmic diversity.

That matters because relying on a single mathematical family creates systemic risk.

18. Security Levels

PQC standards use parameter sets designed to achieve different security strengths.

ECDAT should not simply store:

```
PQC = TRUE
```

It should store detailed information.

Example:

Algorithm:	ML-KEM
Parameter:	ML-KEM-768
Family:	Lattice
Function:	KEM
Security Level:	Standardized security category
Key Size:	...

This metadata enables meaningful recommendations.

19. PQC Compared With RSA

A simplified conceptual comparison:

Property	RSA	ML-KEM
Primary Function	Encryption / Signatures	Key Encapsulation
Mathematical Basis	Integer Factorization	Lattice Problems
Quantum Vulnerability	High	Designed for PQ

ty		threat
Typical Legacy Deployement	Very High	Emerging
Migration Difficulty	Existing ecosystem	Integration required

The critical lesson:

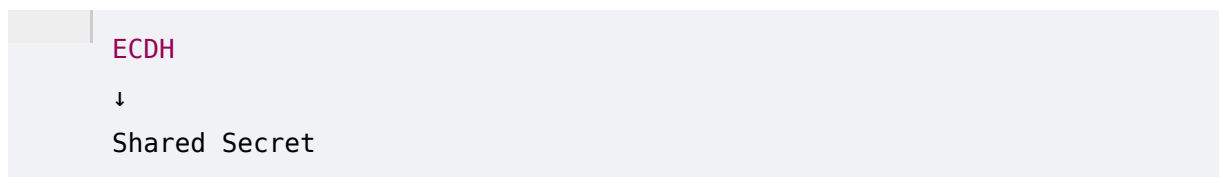
ML-KEM is not a direct replacement for RSA signatures.

If RSA is being used for signatures, a signature scheme such as ML-DSA or SLH-DSA must be considered.

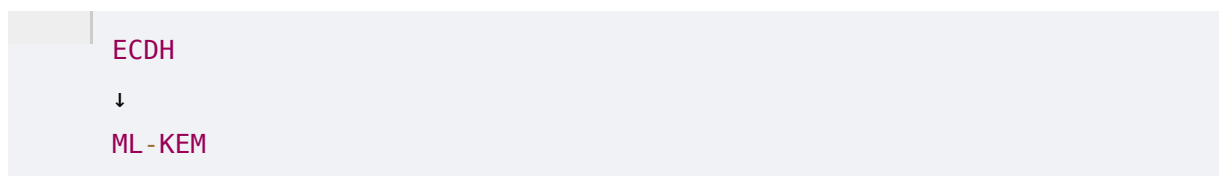
20. PQC Compared With ECC

Consider ECDH.

Its role:



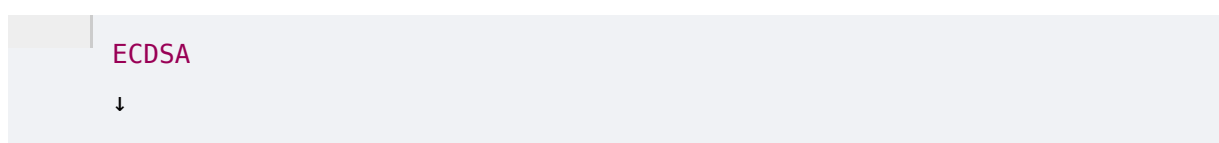
A migration path may involve:



or a hybrid:



For ECDSA:



The recommendation depends on:

- Protocol
- Application
- Hardware
- Compliance
- Performance
- Ecosystem compatibility

This is exactly the kind of reasoning ECDAT must automate.

21. PQC Compared With AES

PQC migration does not mean replacing every cryptographic algorithm.

This is one of the most important points in the project.

An enterprise might have:

AES-256

+

RSA-2048

The quantum migration concern is primarily the public-key component.

The enterprise may retain AES-256 while migrating:

RSA

↓

PQC KEM / signature

This dramatically changes the scope of migration.

ECDAT should therefore avoid generating meaningless recommendations such as:

"Replace AES-256 with PQC."

PQC is primarily relevant to replacing vulnerable public-key mechanisms and supporting appropriate quantum-resistant cryptographic primitives.

22. Hybrid Cryptography

One of the most important migration strategies is:

Hybrid Cryptography

Instead of immediately abandoning classical cryptography:

Classical

an enterprise can use:

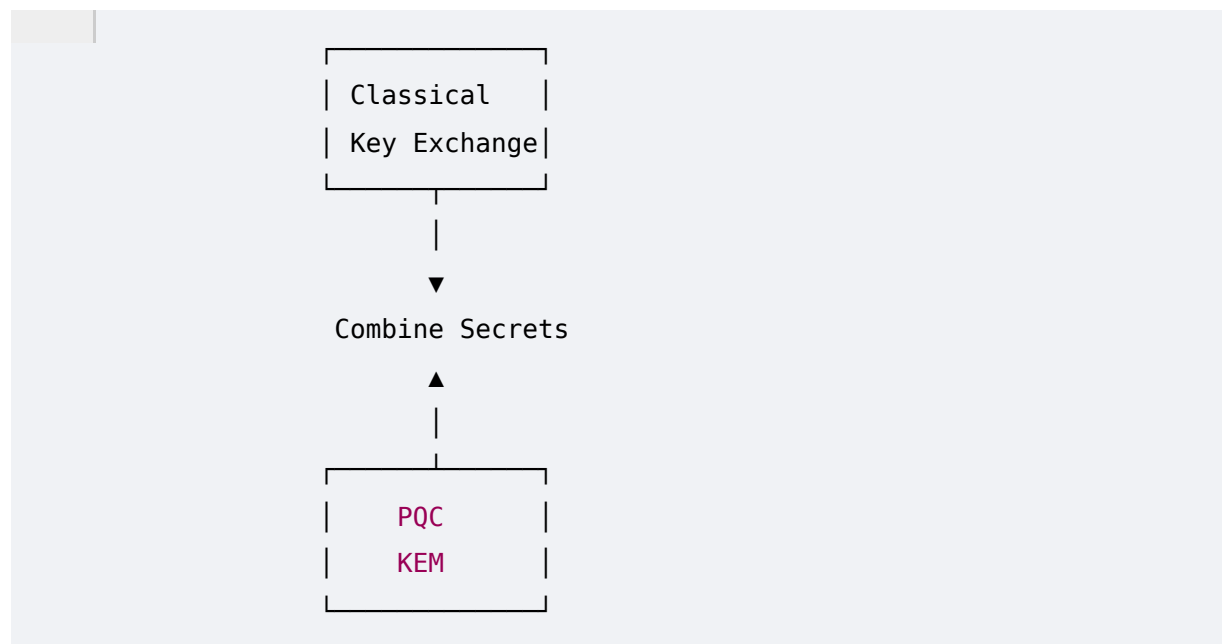
Classical
+
PQC

For example:

ECDH
+
ML-KEM

The resulting key establishment can derive security from both components.

Conceptually:



The exact construction must follow the relevant protocol specification and cryptographic guidance.

23. Why Hybrid Migration Matters

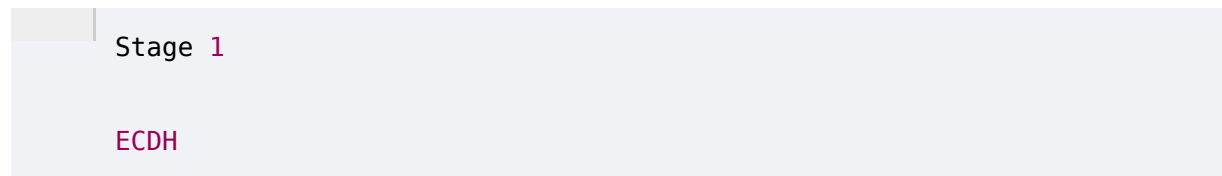
Enterprises may be reluctant to immediately remove classical algorithms.

Reasons include:

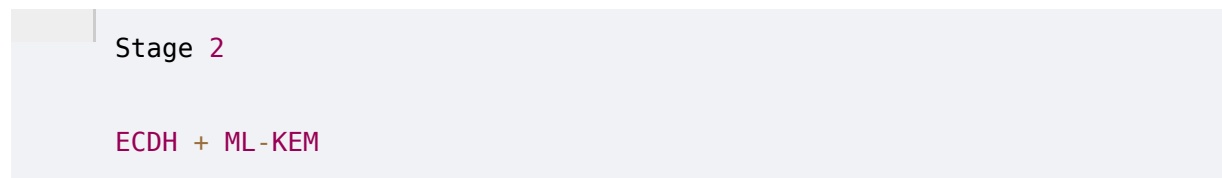
- Legacy clients
- Old hardware
- Compatibility
- Regulatory requirements
- Vendor support
- Unknown implementation risks

Hybrid migration provides a transition strategy.

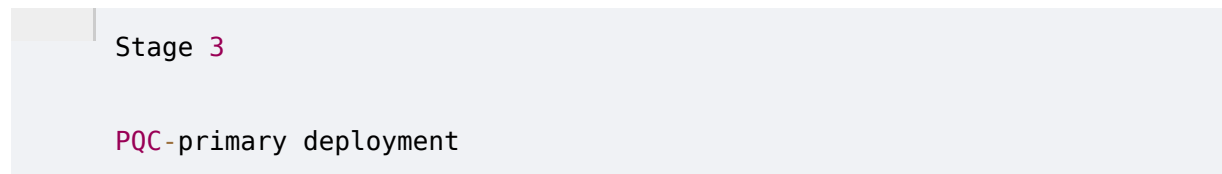
Example:



then:



then:



The exact sequence depends on the protocol and organizational requirements.

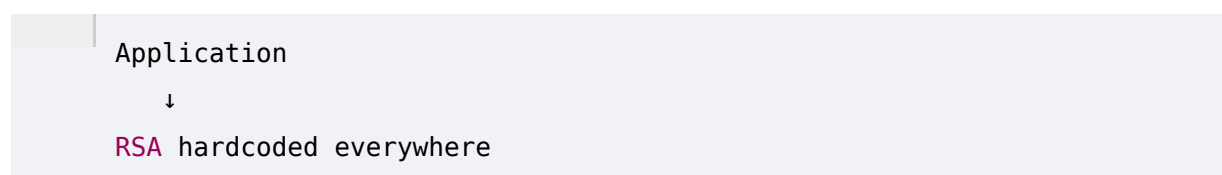
24. Crypto-Agility

A critical concept for ECDAT is:

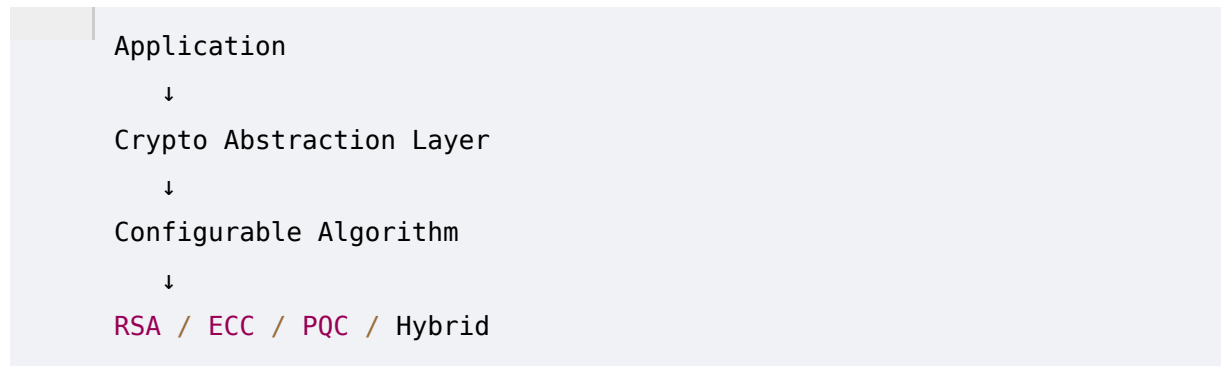
Crypto-Agility

Crypto-agility means an organization's systems can change cryptographic algorithms without rebuilding the entire infrastructure.

Bad architecture:



Good architecture:



Crypto-agility makes future migrations dramatically easier.

25. PQC Migration Challenges

PQC migration is not simply a software update.

Organizations may encounter:

Large keys

Some PQC algorithms use larger keys than classical ECC.

Larger signatures

Some PQC signatures can be substantially larger.

Network overhead

Larger cryptographic messages can affect protocols.

Hardware limitations

HSMs and smart cards may not support PQC immediately.

Embedded systems

Firmware may have extremely long replacement cycles.

Certificate infrastructure

PQC certificate support requires ecosystem changes.

Third-party dependencies

An enterprise cannot always modify vendor software.

26. Performance Considerations

ECDAT should evaluate performance.

Important metrics include:

Key Generation Time

Encapsulation Time

Decapsulation Time

Signing Time

Verification Time

Memory Usage

CPU Usage

Network Overhead

Storage Overhead

Suppose two algorithms provide appropriate security.

One has:

Latency = 2 ms

and another:

Latency = 20 ms

For a latency-sensitive trading platform, this difference could be significant.

Therefore:

Algorithm selection is an engineering optimization problem, not merely a security classification problem.

27. Key Size and Ciphertext Size

PQC algorithms can have larger keys and ciphertexts than traditional elliptic-curve systems.

This matters because larger cryptographic objects affect:

- Network bandwidth

- Certificate sizes
- Storage
- Memory
- Protocol packet sizes
- Hardware buffers

ECDAT should therefore include size metadata.

Example:

```
Algorithm
├─ Public Key Size
├─ Private Key Size
├─ Ciphertext Size
└─ Signature Size
```

28. Signature Size

Digital signatures may be transmitted frequently.

Examples:

- TLS certificates
- Software updates
- Container signatures
- Firmware
- Documents
- Blockchain transactions

Therefore, larger signatures can affect system performance.

ECDAT should analyze:

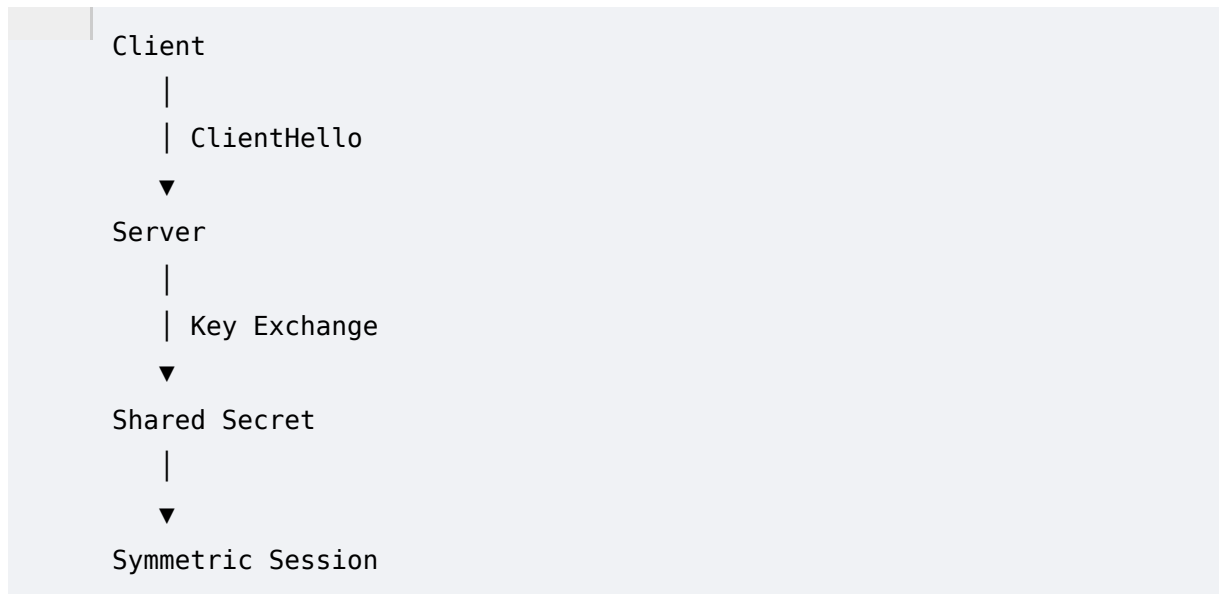
```
Signature Frequency
x
Signature Size
x
Network Constraints
```

when evaluating migration impact.

29. TLS and PQC

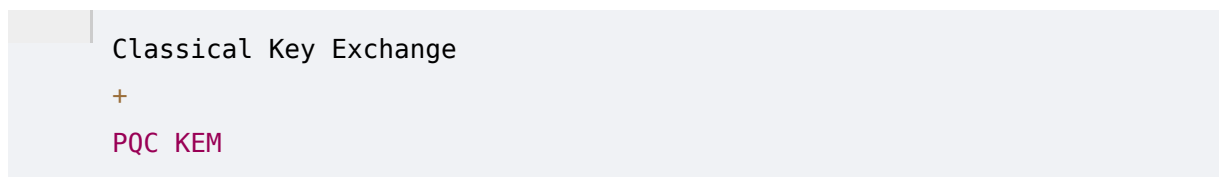
TLS is one of the most important migration targets.

A typical TLS connection involves:



The public-key component is particularly relevant to PQC migration.

A future-compatible architecture may involve:



during a transition period.

ECDAT should identify:

- TLS version
- Cipher suites
- Key exchange mechanism
- Certificate algorithm
- Library
- Library version
- Server configuration

30. Certificates and PQC

Certificates are a major migration challenge.

A traditional certificate may contain:

```
Subject
Public Key
Issuer
Validity
Extensions
Signature
```

If the public key algorithm is RSA or ECC, the certificate ecosystem must eventually support PQC alternatives.

This creates challenges involving:

- Certificate authorities
- Browsers
- Operating systems
- TLS libraries
- HSMs
- Network appliances

ECDAT should therefore treat certificate inventory as a major part of PQC readiness.

31. PKI Migration

PKI is deeply embedded in enterprise infrastructure.

Consider:

```
Root CA
  ↓
Intermediate CA
  ↓
Server Certificate
  ↓
Application
```

Changing the cryptographic algorithm can affect the entire chain.

Therefore, ECDAT should identify:

```
Certificate
  ↓
Issuer
  ↓
Algorithm
```



```
graph TD; A[Application] --> B[Service]; B --> C[Business Owner];
```

This is an excellent use case for a knowledge graph.

32. HSM Migration

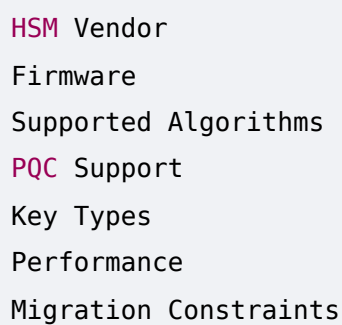
Hardware Security Modules protect cryptographic keys.

They are widely used by:

- Banks
- Governments
- Cloud providers
- Payment systems
- Identity providers

PQC migration requires determining whether existing HSMs support the required algorithms.

ECDAT should track:



- HSM Vendor
- Firmware
- Supported Algorithms
- PQC Support
- Key Types
- Performance
- Migration Constraints

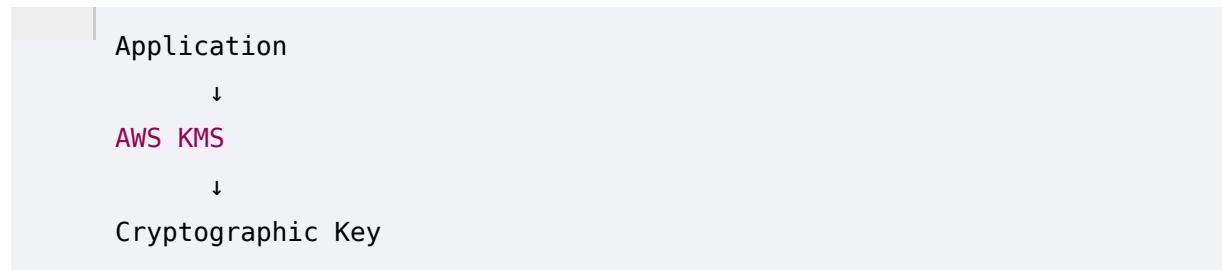
33. Cloud Migration

Modern organizations use cloud services such as:

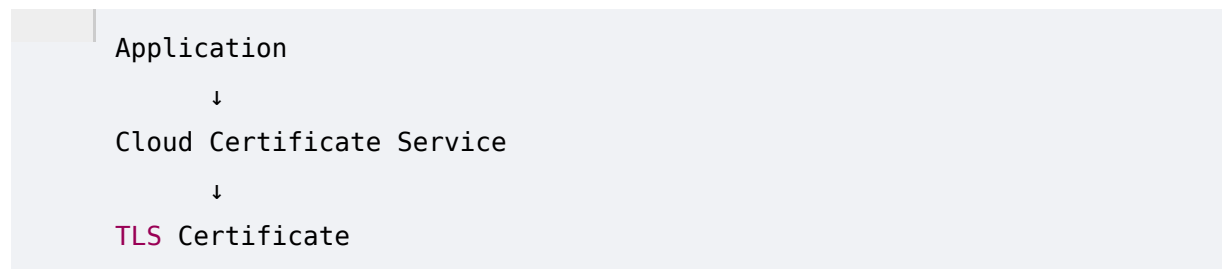
- Key Management Services
- Secrets Managers
- Certificate Managers
- Identity Services
- API Gateways

Cryptographic dependencies may therefore be hidden inside managed services.

Example:



or:



ECDAT should eventually integrate with cloud APIs to discover these dependencies.

34. API and Application Migration

Applications may directly call cryptographic APIs.

Example:

```
RSA.generate()
```

or:

```
ECDSA.sign()
```

But the harder cases look like:

```
SecurityProvider.sign()
```

where the actual algorithm is determined by configuration.

Therefore, ECDAT requires both:

Syntactic analysis

Find explicit cryptographic APIs.

Semantic analysis

Understand what the application actually does.

This distinction becomes extremely important in the AI phase.

35. Embedded Systems

Embedded systems create one of the biggest PQC migration challenges.

Examples:

- Cars
- Industrial controllers
- Satellites
- Medical devices
- Routers
- IoT devices
- Hardware appliances

Some devices may remain deployed for:

10–20+ years

If cryptographic algorithms are embedded directly in firmware, replacement may require:

- Firmware updates
- Hardware replacement
- Certification
- Field deployment
- Physical access

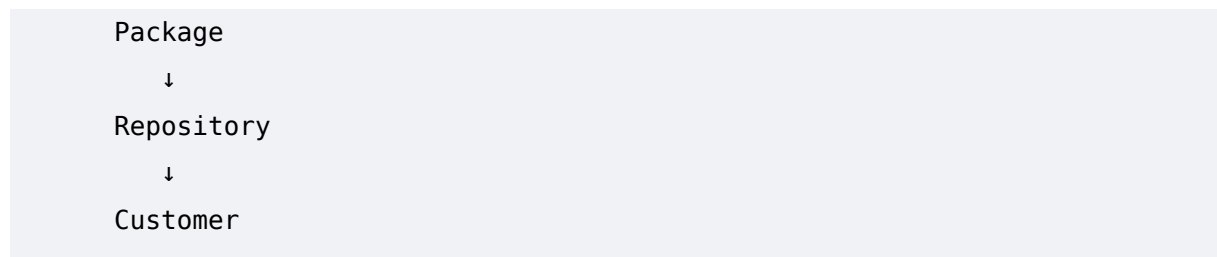
ECDAT should therefore assign higher migration complexity to such assets.

36. Software Supply Chain

Software is often signed.

For example:

```
Developer
  ↓
Build System
  ↓
Code Signing
  ↓
```



If the signing algorithm is quantum-vulnerable, future attackers could potentially target the trust chain.

ECDAT should discover:

- Code-signing keys
- Signing algorithms
- Package signatures
- Certificate chains
- Build infrastructure
- Artifact repositories

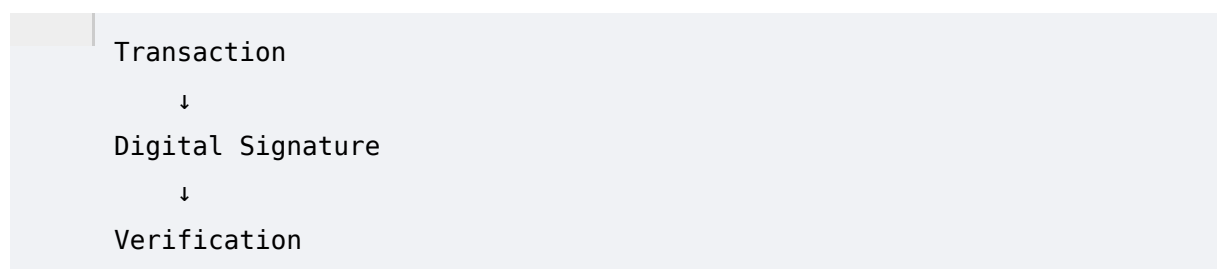
37. Blockchain and PQC

The SIH theme includes:

Blockchain & Cybersecurity

Blockchain systems are particularly interesting because many rely on digital signatures.

For example:



If a blockchain's signature scheme is vulnerable to quantum attacks, an attacker with sufficient quantum capability could potentially threaten authentication mechanisms.

PQC blockchain migration can therefore involve:

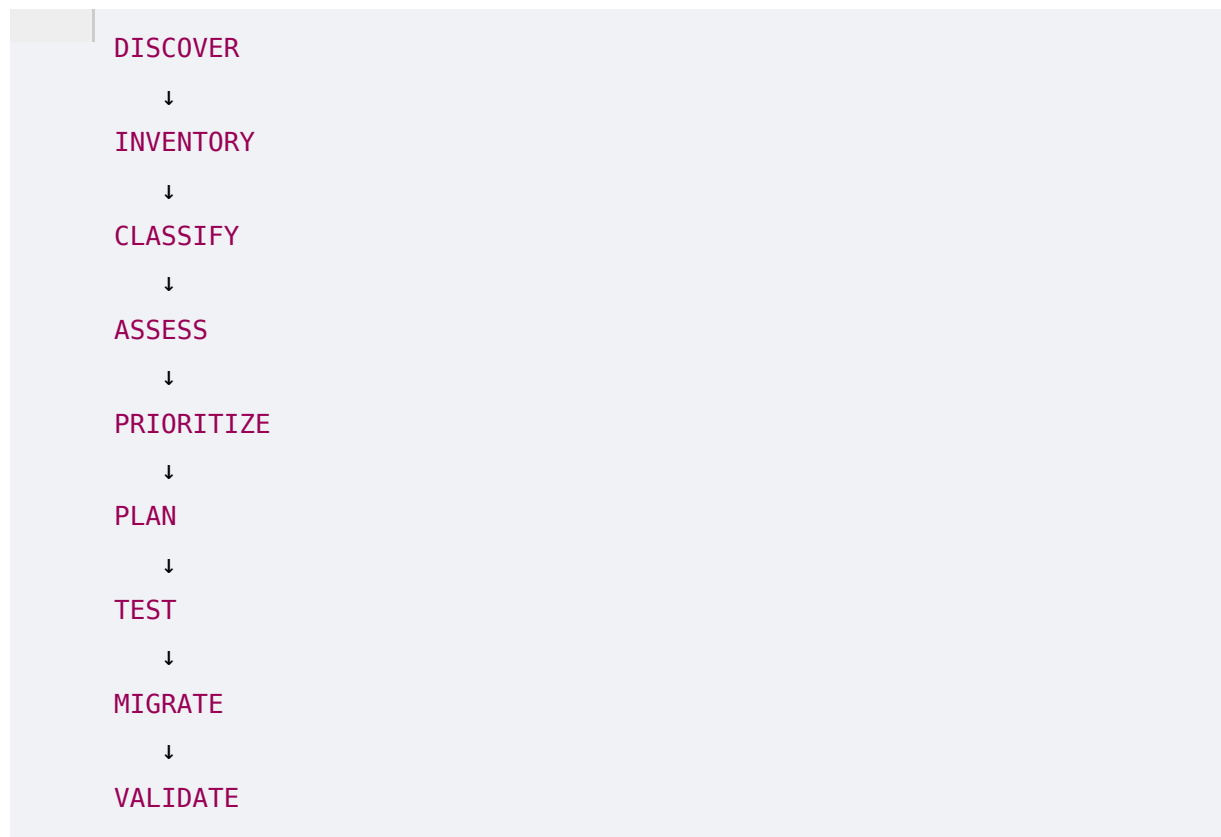
- Quantum-resistant signatures
- Wallet migration
- Address migration

- Smart contract changes
- Consensus implications

This is a specialized future extension for ECDAT.

38. Enterprise PQC Migration

A mature enterprise migration can be represented as:



This should become ECDAT's core lifecycle.

39. PQC Discovery Requirements for ECDAT

ECDAT should detect at least:

Algorithms

- RSA
- DH
- ECDH
- ECDSA
- EdDSA
- AES
- SHA family

- Other cryptographic primitives

PQC

- ML-KEM
- ML-DSA
- SLH-DSA
- Hybrid mechanisms

Context

- Protocol
 - Library
 - Version
 - Application
 - Repository
 - Container
 - Certificate
 - Key
 - Cloud service
-

40. PQC Recommendation Engine

This is where ECDAT becomes more than a scanner.

Suppose it discovers:

ECDSA P-256

The recommendation engine should ask:

What is the purpose?

Signature?

Certificate?

Code signing?

API authentication?

Firmware?

Then:

What is the business criticality?

What is the performance requirement?

What hardware supports it?

What protocol is being used?

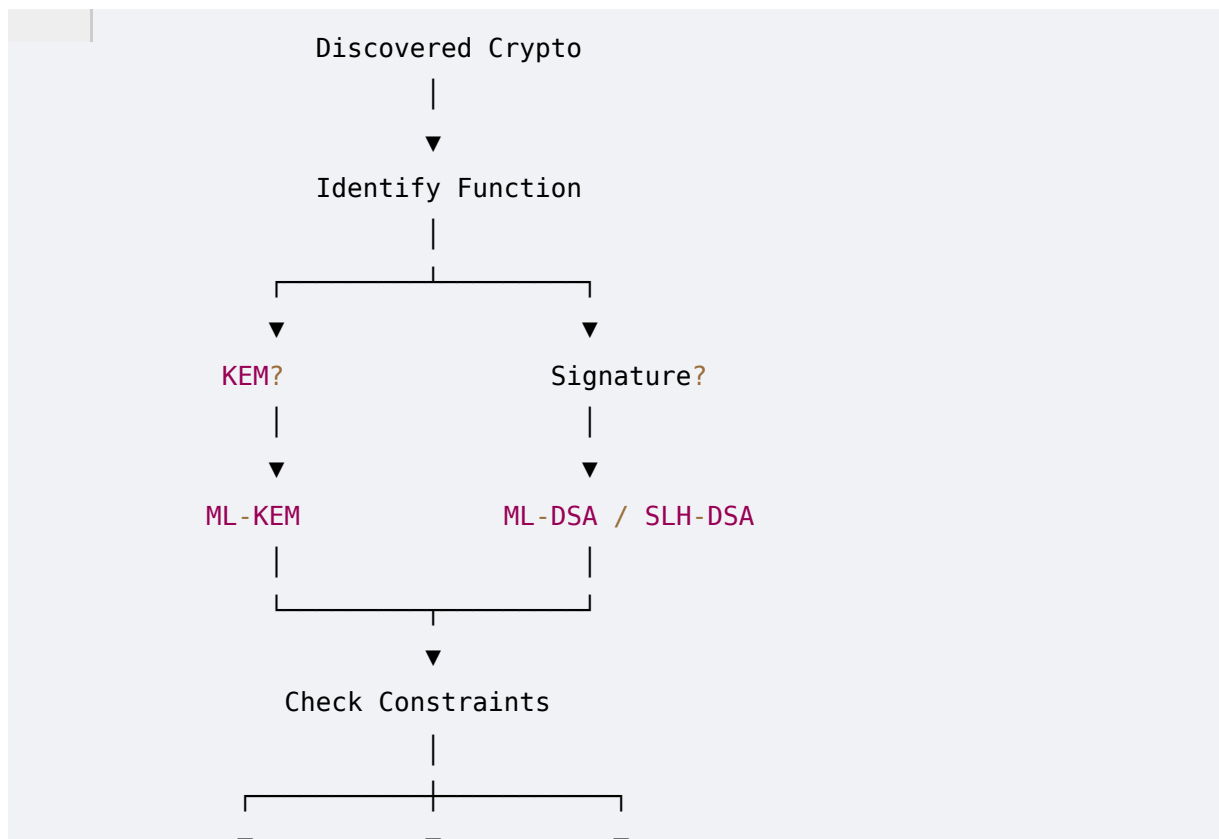
Does the ecosystem support PQC?

Is hybrid deployment required?

Only then should it recommend an algorithm.

41. Algorithm Selection Framework

A possible decision process:



42. Migration Decision Matrix

ECDAT could produce a matrix like:

Existing	Function	Potential Direction	Strategy
RSA	Key establishment	ML-KEM	Hybrid / PQC
ECDH	Key establishment	ML-KEM	Hybrid / PQC
RSA	Signature	ML-DSA / SLH-DSA	Hybrid / PQC
ECDSA	Signature	ML-DSA / SLH-DSA	Hybrid / PQC
AES-128	Symmetric	AES-256 consideration	Parameter review
SHA-1	Hash	SHA-2/SHA-3	Immediate classical remediation
3DES	Symmetric	AES-family alternatives	Immediate remediation

This table must be treated as a conceptual framework, not an automatic replacement map.

Actual migration decisions require protocol and implementation context.

43. Example ECDAT Recommendation

Suppose ECDAT finds:

Asset:
PAYMENT-GATEWAY-01

Algorithm:
ECDSA P-256

Purpose:
TLS Certificate

Business Criticality:
Critical

Traffic:
High

Migration Complexity:
High

ECDAT might generate:

QUANTUM RISK:
HIGH

REASON:
ECDSA is based on elliptic-curve discrete logarithms, which are vulnerable to sufficiently capable quantum computers using Shor's algorithm.

RECOMMENDATION:
Begin migration planning toward a standardized post-quantum signature mechanism.

TRANSITION:
Evaluate hybrid certificate / protocol deployment where supported.

DEPENDENCIES:
Certificate Authority

This is the type of report NTRO's expected solution implies.

44. PQC Risk Categories

ECDAT can categorize migration readiness.

Category 1 — Quantum Ready

- PQC deployed
- + Crypto-agile
- + Inventory complete

Category 2 — Migration Ready

- PQC supported
- + Inventory available
- + Migration plan exists

Category 3 — Migration Required

- Quantum-vulnerable crypto
- + Long-lived data
- + Migration not started

Category 4 — Critical

- Quantum-vulnerable crypto
- + Highly sensitive data
- + Long retention
- + High migration complexity

45. PQC Migration Roadmap

A realistic enterprise roadmap:

Stage 1 — Discover

Inventory all cryptographic assets.

Stage 2 — Map

Connect assets to:

- Applications
 - Data
 - Owners
 - Infrastructure
-

Stage 3 — Assess

Determine:

- Quantum vulnerability
 - Business impact
 - Data lifetime
 - Migration complexity
-

Stage 4 — Prioritize

Use Mosca-style analysis and organizational priorities.

Stage 5 — Pilot

Deploy PQC in controlled environments.

Stage 6 — Hybrid

Deploy classical + PQC where appropriate and standardized.

Stage 7 — Migrate

Move production systems.

Stage 8 — Decommission

Retire vulnerable legacy cryptography.

Stage 9 — Continuous Monitoring

Continue scanning for newly introduced vulnerable cryptography.

46. Common PQC Misconceptions

"PQC means encryption becomes quantum."

No.

PQC runs on conventional computers.

"Every cryptographic algorithm needs replacement."

No.

The primary urgency is around cryptographic constructions whose security is threatened by quantum algorithms.

"PQC means ML-KEM everywhere."

No.

ML-KEM is a KEM.

Signatures require signature algorithms.

"PQC is automatically secure."

No.

PQC algorithms depend on mathematical assumptions and implementations that must continue to be evaluated.

"Hybrid means just concatenate two algorithms."

No.

Hybrid cryptography requires carefully specified composition methods.

"Once PQC is deployed, migration is finished."

No.

Organizations need continuing crypto-agility and inventory.

47. Engineering Requirements

Phase 3 produces important engineering requirements for ECDAT.

Requirement A

Maintain a **PQC algorithm registry**.

Requirement B

Track:

Algorithm
Family
Function
Parameter Set
Security Level
Quantum Status
Classical Status
Key Size
Signature Size
Ciphertext Size
Performance
Recommended Uses

Requirement C

Maintain algorithm lifecycle state:

Experimental
Candidate
Standardized
Recommended
Deprecated
Broken

Requirement D

Support migration relationships:

RSA
↓
Potential Alternatives

ECDSA

↓

Potential Alternatives

ECDH

↓

Potential Alternatives

Requirement E

Support hybrid strategies.

Requirement F

Allow administrators to configure organizational policies.

Example:

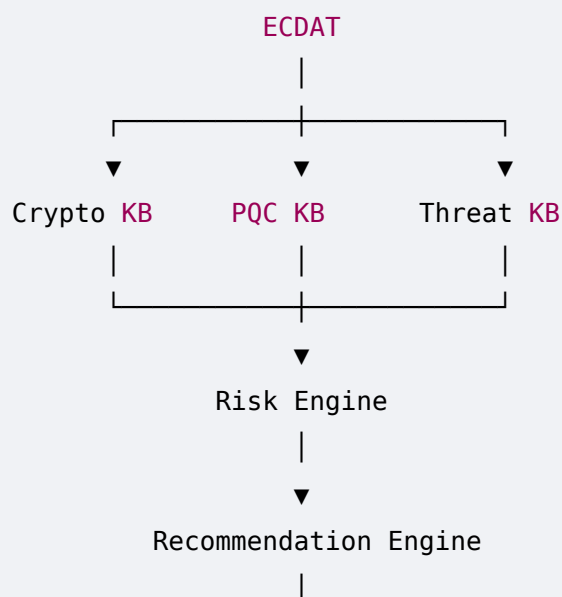
Policy:

No newly deployed RSA
after 2028.

All critical systems must
support PQC migration planning.

48. Phase 3 Architecture Implications

By now, ECDAT requires several knowledge systems.

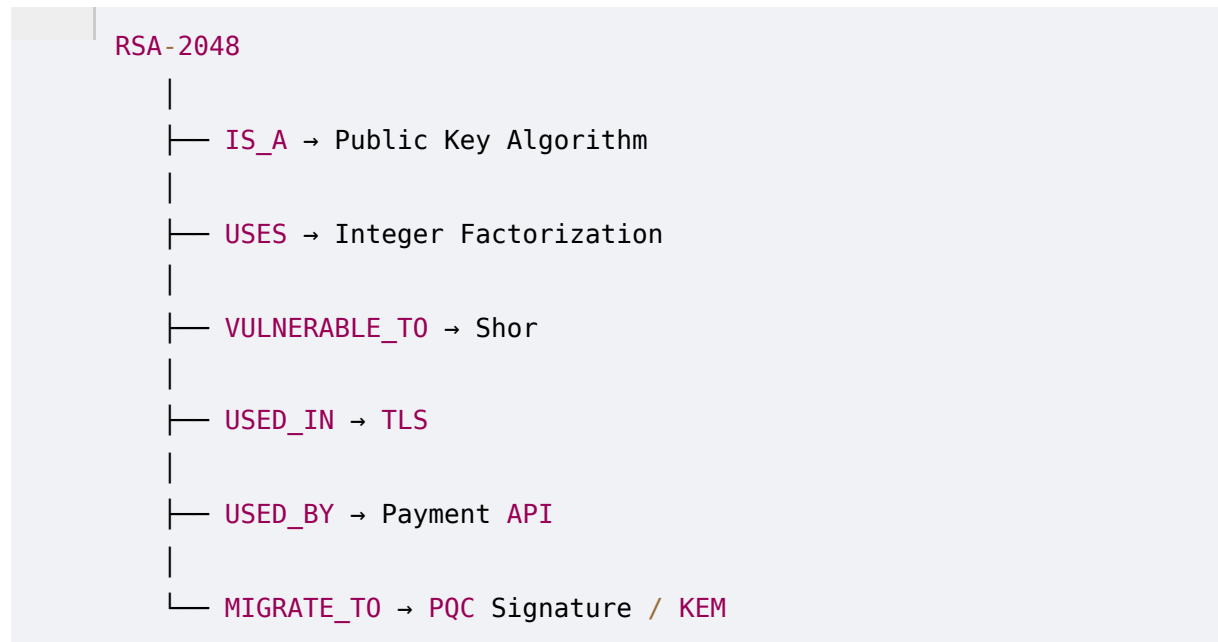


▼ Migration Plan

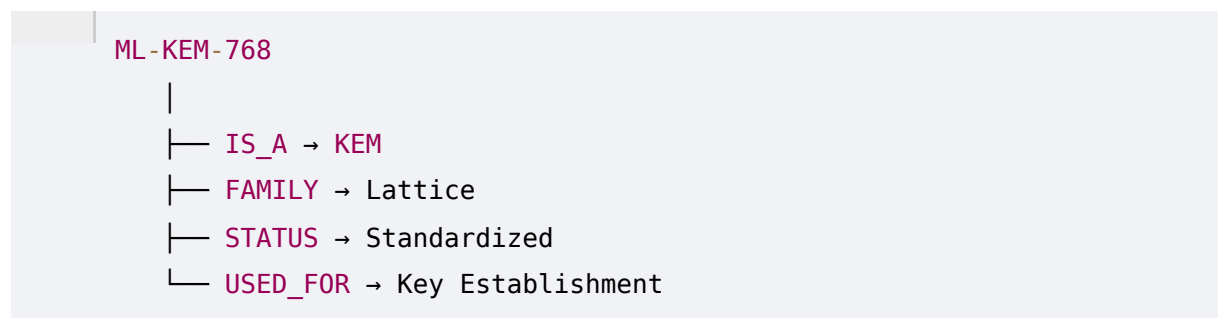
This knowledge layer will eventually become one of the most important components of the platform.

49. ECDAT's PQC Knowledge Graph

A cryptographic knowledge graph could represent:



Similarly:



This graph enables much more powerful queries than a traditional relational inventory.

50. Example Knowledge Query

A security administrator could ask:

"Show all critical applications using quantum-vulnerable public-key cryptography that protect data with a retention period greater than 10 years."

ECDAT could translate that into a graph query conceptually equivalent to:

```
Application
→ uses
Cryptographic Asset
→ algorithm
Quantum Vulnerable
→ protects
Data
→ retention > 10 years
→ criticality = critical
```

The system returns a prioritized migration list.

This is where **PQC knowledge + CBOM + Knowledge Graph + AI** begin to converge.

51. Phase 3 Summary

Post-Quantum Cryptography represents the next generation of cryptographic mechanisms intended to withstand attacks from sufficiently capable quantum computers.

The most important enterprise concepts are:

Key Encapsulation

ML-KEM

Used for establishing shared secrets.

Digital Signatures

ML-DSA

A lattice-based standardized signature mechanism.

SLH-DSA

A hash-based standardized signature mechanism.

Migration Strategy

Hybrid cryptography

Combines classical and post-quantum mechanisms during transition where appropriate.

Architectural Strategy

Crypto-agility

Allows organizations to change cryptographic algorithms without massive system redesign.

52. The Most Important Insight

The goal of ECDAT is **not**:

"Find RSA and tell the user to use ML-KEM."

That would be far too simplistic.

The real goal is:

Understand the cryptographic function, the implementation, the protocol, the data, the business context, the migration constraints, and the organization's quantum threat model, then recommend an appropriate migration path.

That is a fundamentally different problem.

ECDAT is therefore evolving from:

Scanner

into:

Cryptographic Intelligence Platform

and ultimately:

Enterprise Quantum Readiness Platform

53. Phase 3 Core Model

The entire phase can be condensed into:

EXISTING CRYPTOGRAPHY

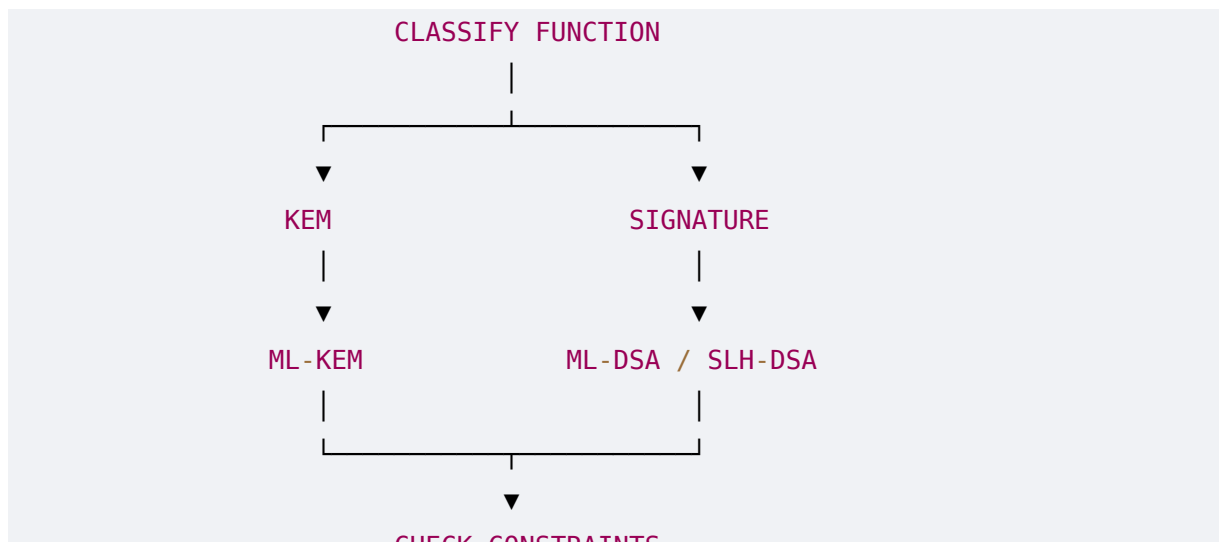
|



DISCOVER & MAP

|





54. Phase 3 Final Takeaway

The post-quantum transition is not simply an algorithm replacement exercise.

It is a transformation of an organization's entire cryptographic infrastructure.

The organization must move from:

Unknown Cryptography

to:

Cryptographic Inventory

then:

Inventory
↓
Risk Assessment
↓
PQC Readiness
↓
Migration Planning
↓
Crypto-Agility

This is precisely why the NTRO problem statement is interesting.

It combines **cryptographic discovery, quantum risk assessment, enterprise architecture, AI-assisted analysis, and migration planning** into a single platform.

55. Phase 4 Preview — Enterprise Cryptographic Ecosystem

Now we have established:

Phase 1: What cryptography is.

Phase 2: Why quantum computing threatens parts of it.

Phase 3: What post-quantum alternatives look like.

The next question is the hardest practical one:

Where exactly does cryptography exist inside a real enterprise?

Phase 4 will map the entire enterprise cryptographic ecosystem, including:

- Source code
- Programming languages
- Cryptographic APIs
- OpenSSL
- BoringSSL
- LibreSSL
- Java JCA/JCE
- .NET cryptography
- Python cryptography
- Go crypto
- RustCrypto
- Node.js crypto
- Native libraries
- Shared libraries
- Static binaries
- Dynamic binaries
- Package managers
- SBOMs
- Containers
- Docker
- Kubernetes
- TLS

- SSH
- VPN
- PKI
- Certificates
- HSMs
- Cloud KMS
- Secrets managers
- Databases
- Object storage
- API gateways
- Service meshes
- CI/CD
- Code signing
- Firmware
- IoT
- Mobile applications
- Embedded systems
- Blockchain
- Third-party SaaS
- Legacy systems

Most importantly, we'll start translating the **NTRO requirement "scan source code repositories, binaries, libraries and container images"** into a concrete technical architecture for ECDAT.

Phase 3 complete.